

Lab 7: Two-Stage Least Squares and Its Diagnostics

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1 Goals

1. Build the IV estimator by hand and see that it is a ratio of two reduced-form quantities.
2. Run 2SLS properly, and understand why running the two stages manually gives the wrong standard errors.

3. Read the diagnostics an IV paper should report: the first-stage F , the reduced form, and the Anderson–Rubin test.
4. See, in simulation, what a weak instrument does to the 2SLS output R prints.
5. Characterize the compliers.

```
library(dplyr)
library(ggplot2)
library(ivreg)          ## ivreg(): 2SLS
library(fixest)          ## feols(): 2SLS with robust SEs and fixed
effects
library(sandwich); library(lmtest)
library(ivDiag)          ## AR_test(): Anderson-Rubin test and interval
library(ivdesc)          ## ivdesc(): complier profiles

## Card's (1995) NLS Young Men extract; log wage and a binary instrument
data("SchoolingReturns", package = "ivreg")
card <- SchoolingReturns |>
  mutate(lwage = log(wage),
         near4 = as.integer(nearcollege4 != "none"),    ## grew up near a 4-year
         college
         age2 = age^2,
         black = as.integer(ethnicity == "afam"),
         smsa_yes = as.integer(smsa == "yes"),
         south_yes = as.integer(south == "yes"))
c(n = nrow(card), share_near_4yr = mean(card$near4))
#>           n share_near_4yr
#>    3010.0000      0.6821
table(card$nearcollege4)
#>
#>   none  public private
#>   957   1483    570
```

2 The design

The question is the causal return to a year of schooling. Schooling is endogenous: unmeasured ability raises both schooling and wages, so OLS is biased upward; measurement error pushes the other way.

The instrument, following Card (1995), is **growing up near a four-year college**. The argument for relevance is that proximity lowers the cost of attending. The argument for exclusion is that, conditional on region and urban status, where your parents happened to live does not affect your wage except through the schooling it made cheaper.

Warning

That exclusion argument is contestable, and it has been contested — college towns differ from other places in labour markets, school quality, and family background. Part of today's job is learning to interrogate it rather than accept it.

3 By hand: the Wald estimator

With a binary instrument, IV is the ratio of two differences in means.

```
## reduced form: the instrument's effect on the outcome
rf <- mean(card$lwage[card$near4 == 1]) - mean(card$lwage[card$near4 == 0])
## first stage: the instrument's effect on the treatment
fs <- mean(card$education[card$near4 == 1]) - mean(card$education[card$near4 == 0])
```

```
c(reduced_form = rf, first_stage = fs, wald_ratio = rf / fs)
#> reduced_form first_stage wald_ratio
#>          0.1559         0.8290         0.1881
```

```
## the same numbers as regressions
c(reduced_form = coef(lm(lwage ~ near4, data = card))["near4"],
  first_stage = coef(lm(education ~ near4, data = card))["near4"],
  iv          = coef(ivreg(lwage ~ education | near4, data = card))["education"])
#> reduced_form.near4 first_stage.near4 iv.education
#>          0.1559         0.8290         0.1881
```

Note

Notice the two ingredients. The reduced form is small (about 0.16 log points) and so is the first stage (about 0.83 years). Dividing one small number by another is exactly the operation that makes IV fragile — and once we add covariates, both shrink further.

4 2SLS with covariates

The controls, used throughout the lab, are age and its square, race, living in a metropolitan area, and living in the South. Age is fixed at birth, so it is exogenous. We do **not** use potential experience: it is computed from schooling ($\text{exper} = \text{age} - \text{educ} - 6$), so it would be endogenous too and would need an instrument of its own. With age, the model has one endogenous regressor (education) and one excluded instrument (near4): it is just identified.

```
## OLS with the controls
m_ols <- lm(lwage ~ education + age + I(age^2) + ethnicity + smsa + south,
            data = card)

## 2SLS: outcome ~ regressors | instruments
## the exogenous controls appear on BOTH sides of the bar;
## near4 replaces education on the right
m_iv <- ivreg(lwage ~ education + age + I(age^2) + ethnicity + smsa + south |
              near4 + age + I(age^2) + ethnicity + smsa + south,
              data = card)

round(cbind(OLS = coef(m_ols)[1:2], IV = coef(m_iv)[1:2]), 4)
#>               OLS      IV
#> (Intercept) 3.1906 3.2757
#> education   0.0341 0.0936
```

```
## heteroskedasticity-robust standard errors; diagnostics are read in section
5
summary(m_iv, vcov = sandwich, diagnostics = FALSE)
#>
#> Call:
#> ivreg(formula = lwage ~ education + age + I(age^2) + ethnicity +
#>       smsa + south | near4 + age + I(age^2) + ethnicity + smsa +
#>       south, data = card)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -1.8594 -0.2522  0.0226  0.2581  1.3111
#>
#> Coefficients:
#>               Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   3.275667   0.698537   4.69 0.0000029 ***
#> education     0.093607   0.049060   1.91  0.05648 .
#> age           0.082440   0.070274   1.17  0.24084
#> I(age^2)     -0.000733   0.001227  -0.60  0.55039
#> ethnicityafam -0.101177   0.073367  -1.38  0.16798
#> smsayes       0.108067   0.049642   2.18  0.02957 *
#> southyes      -0.099427   0.029796  -3.34  0.00086 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 0.406 on 3003 degrees of freedom
```

```
#> Multiple R-Squared: 0.166,   Adjusted R-squared: 0.164
#> Wald test: 162 on 6 and 3003 DF, p-value: <0.0000000000000002
```

The IV estimate is *larger* than OLS. That is the standard finding with this instrument and it is worth pausing on: if ability bias were the only problem, IV should come out *smaller*. Candidate explanations are measurement error in self-reported schooling, and the fact that the LATE applies to compliers whose return may genuinely be higher.

4.1 Why not run the two stages by hand?

```
## stage 1: education on the instrument and the controls
stage1 <- lm(education ~ near4 + age + I(age^2) + ethnicity + smsa + south,
             data = card)
card$edu_hat <- fitted(stage1)
## stage 2: the outcome on the fitted education and the same controls
stage2 <- lm(lwage ~ edu_hat + age + I(age^2) + ethnicity + smsa + south,
             data = card)

## same coefficient, different standard error
round(cbind(
  coefficient = c(manual = coef(stage2)["edu_hat"], ivreg =
    coef(m_iv)["education"]),
  std_error = c(manual = sqrt(diag(vcov(stage2)))["edu_hat"],
    ivreg = sqrt(diag(vcov(m_iv)))["education"])
), 5)
#>               coefficient std_error
#> manual.edu_hat      0.09361    0.04734
#> ivreg.education     0.09361    0.04971
```

The **coefficients agree** but the **standard errors do not**. The manual second stage computes its residuals with $\hat{D}$ instead of the actual D , so it estimates the wrong error variance. Always use `ivreg()` or `feols()`.

4.2 The same thing in fixest

```
## feols syntax: outcome ~ controls | endogenous ~ instrument
m_fe <- feols(lwage ~ age + I(age^2) + ethnicity + smsa + south |
             education ~ near4,
             data = card, vcov = "hetero")
## fitstat = ~ ivwald: show only the robust first-stage test under the table
print(m_fe, fitstat = ~ ivwald)
#> TSLS estimation: Second stage
```

```

#> |- D.V.      : lwage
#> |- Endo.     : education
#> |- Instr.    : near4
#> Dep. Var.: lwage
#> Observations: 3,010
#> Standard-errors: Heteroskedasticity-robust
#>
#>           Estimate Std. Error t value      Pr(>|t|)
#> (Intercept)   3.275667   0.699350   4.6839 0.0000029401 ***
#> fit_education  0.093607   0.049117   1.9058 0.0567710800 .
#> age           0.082440   0.070355   1.1718 0.2413858586
#> I(I(age^2))   -0.000733   0.001229  -0.5966 0.5508518552
#> ethnicityafam -0.101177   0.073452  -1.3775 0.1684746903
#> smsayes       0.108067   0.049700   2.1744 0.0297549961 *
#> southyes      -0.099427   0.029831  -3.3330 0.0008694188 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#> Wald (1st stage), education: stat = 10.224, p = 0.001401, on 1 and 3,003
DoF, VCOV: Heteroskedasticity-robust.

```

```

## two versions of the first-stage test of the instrument
fitstat(m_fe, c("ivf", "ivwald"))
#> F-test (1st stage), education: stat = 10.524, p = 0.001191, on 1 and 3,003
DoF.
#> Wald (1st stage), education : stat = 10.224, p = 0.001401, on 1 and 3,003
DoF, VCOV: Heteroskedasticity-robust.

```

How to read the two lines:

- **F-test (1st stage)** (`ivf`) tests whether the excluded instrument's coefficient in the first stage is zero, using the classical formula that assumes homoskedastic errors.
- **Wald (1st stage)** (`ivwald`) tests the same hypothesis with the model's own variance estimate, here heteroskedasticity-robust. With one instrument it is the squared robust t of `near4` in the first stage. Report this one when your standard errors are robust or clustered.

4.3 Several endogenous regressors

When a model has more than one endogenous regressor, it needs at least as many excluded instruments. In `feols()`, list the endogenous regressors on the left of `~` in the last part of the formula and the instruments on the right. The order in either list does not matter: 2SLS does not pair an instrument with a regressor. Every endogenous regressor is regressed on *all* the instruments (and the controls), one first stage each, and the second stage uses all the fitted values together. What matters is that the instruments move the regressors in different directions. Card's original specification is the classic example: it controls for potential experience, which is

endogenous because it is built from schooling, and instruments experience and its square with age and its square.

```
## syntax: outcome ~ exogenous controls | endogenous1 + endogenous2 ~ z1 + z2
card <- card |> mutate(exper2 = experience^2)
m_two <- feols(lwage ~ ethnicity + smsa + south |
               education + experience + exper2 ~ near4 + age + age2,
               data = card, vcov = "hetero")
print(m_two, fitstat = ~ ivwald)    ## second stage
#> TSLS estimation: Second stage
#> |- D.V.      : lwage
#> |- Endo.     : education, experience, exper2
#> |- Instr.    : near4, age, age2
#> Dep. Var.: lwage
#> Observations: 3,010
#> Standard-errors: Heteroskedasticity-robust
#>
#>           Estimate Std. Error t value      Pr(>|t|)
#> (Intercept)    4.065667   0.599705   6.7794 0.000000000014466 ***
#> fit_education    0.132947   0.050709   2.6218 0.008790993160210 **
#> fit_experience    0.055961   0.025899   2.1608 0.030790969566756 *
#> fit_exper2     -0.000796   0.001328  -0.5992 0.549080414747949
#> ethnicityafam  -0.103140   0.075424  -1.3675 0.171576806794925
#> smsayes         0.107985   0.049387   2.1865 0.028857326264491 *
#> southyes       -0.098175   0.028433  -3.4528 0.000562434711862 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#> Wald (1st stage), education : stat =      8.2155, p = 1.917e-5, on 3 and
3,003 DoF, VCOV: Heteroskedasticity-robust.
#> Wald (1st stage), experience: stat = 1,581.0116, p < 2.2e-16 , on 3 and
3,003 DoF, VCOV: Heteroskedasticity-robust.
#> Wald (1st stage), exper2    : stat = 1,111.6228, p < 2.2e-16 , on 3 and
3,003 DoF, VCOV: Heteroskedasticity-robust.
summary(m_two, stage = 1)    ## one first stage per endogenous regressor
#> IV: First stage: education
#> TSLS estimation: First stage
#> |- D.V.      : lwage
#> |- Endo.     : education, experience, exper2
#> |- Instr.    : near4, age, age2
#> Dep. Var.: education
#> Observations: 3,010
#> Standard-errors: Heteroskedasticity-robust
#>
#>           Estimate Std. Error t value      Pr(>|t|)
#> (Intercept)   -1.86952   4.212461  -0.4438 0.65721325445855017
```

```

#> near4          0.34710    0.108558    3.1974 0.00140102810099310 **
#> age            1.06144    0.296357    3.5816 0.00034687654309216 ***
#> age2           -0.01876    0.005162   -3.6342 0.00028352031810824 ***
#> ethnicityafam -1.46837    0.115388  -12.7255          < 2.2e-16 ***
#> smsayes        0.83540    0.112369    7.4344 0.000000000000013622 ***
#> southyes       -0.45970    0.103011   -4.4626 0.00000839321651449 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#> RMSE: 2.51286   Adj. R2: 0.116756
#> F-test (1st stage): stat = 8.0085, p = 2.579e-5, on 3 and 3,003 DoF.
#>
#> IV: First stage: experience
#> TSLS estimation: First stage
#> |- D.V.      : lwage
#> |- Endo.     : education, experience, exper2
#> |- Instr.    : near4, age, age2
#> Dep. Var.: experience
#> Observations: 3,010
#> Standard-errors: Heteroskedasticity-robust
#>
#>          Estimate Std. Error t value          Pr(>|t|)
#> (Intercept) -4.13048    4.212461 -0.9805 0.32689987292517841
#> near4        -0.34710    0.108558 -3.1974 0.00140102810097933 **
#> age          -0.06144    0.296357 -0.2073 0.83577306506466897
#> age2         0.01876    0.005162  3.6342 0.00028352031809052 ***
#> ethnicityafam 1.46837    0.115388 12.7255          < 2.2e-16 ***
#> smsayes      -0.83540    0.112369 -7.4344 0.000000000000013622 ***
#> southyes      0.45970    0.103011  4.4626 0.00000839321651471 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#> RMSE: 2.51286   Adj. R2: 0.631024
#> F-test (1st stage): stat = 1,612.7, p < 2.2e-16, on 3 and 3,003 DoF.
#>
#> IV: First stage: exper2
#> TSLS estimation: First stage
#> |- D.V.      : lwage
#> |- Endo.     : education, experience, exper2
#> |- Instr.    : near4, age, age2
#> Dep. Var.: exper2
#> Observations: 3,010
#> Standard-errors: Heteroskedasticity-robust
#>
#>          Estimate Std. Error t value          Pr(>|t|)
#> (Intercept)   609.943    91.2114   6.687 0.00000000000270344 ***

```



```

#> near4          -7.251      2.2987   -3.154 0.0016246030093835 **
#> age            -55.452     6.5473   -8.469      < 2.2e-16 ***
#> age2             1.313     0.1164   11.284      < 2.2e-16 ***
#> ethnicityafam   28.206     2.6195   10.768      < 2.2e-16 ***
#> smsayes        -17.554     2.4615   -7.132 0.00000000000012374 ***
#> southyes        11.177     2.1817    5.123 0.0000003197413902 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#> RMSE: 52.7   Adj. R2: 0.610575
#> F-test (1st stage): stat = 1,473.1, p < 2.2e-16, on 3 and 3,003 DoF.

```

Notice in the first stages that `near4`, `age`, and `age2` all appear in each one: the education first stage is not “education on `near4`”.

With several endogenous regressors, a strong first-stage F for each one separately does not guarantee the instruments can tell them apart; papers report the Sanderson–Windmeijer conditional F instead. The rest of this lab stays with one endogenous regressor.

5 Diagnostics

5.1 The first stage

```

## the first stage with the same controls as the second stage
fs_mod <- lm(education ~ near4 + age + I(age^2) + ethnicity + smsa + south,
             data = card)
round(coef(summary(fs_mod))["near4", ], 4)
#> Estimate Std. Error t value Pr(>|t|)
#>      0.3471      0.1070      3.2441      0.0012

## robust F on the excluded instrument: the weak-instrument test
car::linearHypothesis(fs_mod, "near4 = 0", vcov. = vcovHC(fs_mod, "HC1"))
#>
#> Linear hypothesis test:
#> near4 = 0
#>
#> Model 1: restricted model
#> Model 2: education ~ near4 + age + I(age^2) + ethnicity + smsa + south
#>
#> Note: Coefficient covariance matrix supplied.
#>
#>   Res.Df Df    F Pr(>F)
#> 1    3004
#> 2    3003  1 10.2 0.0014 **

```

```
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Growing up near a four-year college is associated with about a third of a year more schooling once covariates are included, and the robust F is right at the traditional threshold of 10.

Note

Reading the weak-instrument test. `ivreg`'s `summary(..., diagnostics = TRUE)` labels it “Weak instruments”; `fixest` labels it “F-test (1st stage)” or “Wald (1st stage)”. All three are the same idea: an F test that the excluded instruments have zero coefficients in the first stage. With one instrument, $F = t^2$.

- **Read the F , not the p -value.** The p -value tests whether the instrument is *irrelevant*, not whether it is *strong enough*. Here $p = 0.001$, and the instrument is still weak.
- The rule of thumb is $F > 10$ (Staiger and Stock 1997). The conventional t -test has its stated size only when $F > 104$ (Lee et al. 2022).

`summary(..., diagnostics = TRUE)` also prints two other tests. **Wu–Hausman** asks whether the 2SLS and OLS estimates differ by more than chance (a test of endogeneity). **Sargan** asks whether several instruments agree with each other; it needs more instruments than endogenous regressors, so it is `NA` in a just-identified model. Neither is part of this course, and we do not use them.

That F is **not reassuring**. Recall from lecture that the threshold of 10 is now known to be far too lax: Lee, McCrary, Moreira, and Porter (2022) show that a conventional 5% t -test on the 2SLS estimate has correct size only when the first-stage F exceeds roughly **104**. At $F \approx 10$ the conventional standard error cannot be trusted, which is why we test with Anderson–Rubin below.

5.2 The reduced form

```
## the outcome on the instrument and the controls, robust SE
rf_mod <- lm(lwage ~ near4 + age + I(age^2) + ethnicity + smsa + south,
            data = card)
round(coeftest(rf_mod, vcov = vcovHC(rf_mod, "HC1"))["near4", ], 4)
#>   Estimate Std. Error   t value   Pr(>|t|)
#>    0.0325    0.0160    2.0326    0.0422
```

Tip

Always report the reduced form. It is the one regression in an IV design that requires no instrument-strength assumption at all. Here the adjusted reduced-form coefficient is about 0.032 with a standard error of 0.016 — a t of roughly 2. That is the entire evidentiary

basis for the IV estimate, and it is thin. If the instrument has no detectable effect on the outcome, the IV estimate is a small number divided by another small number, and its apparent precision is an artefact.

5.3 Anderson–Rubin

The lecture’s four steps: guess β_0 , regress $Y - \beta_0 D$ on the instrument and the controls, test the instrument’s coefficient, and keep every guess that is not rejected. `ivDiag::AR_test()` runs this test over a grid of guesses. It has no standard-error option and prints no standard error: each test on the grid uses heteroskedasticity-robust standard errors internally (clustered ones if you add `cl = "your_cluster_variable"`), and the output is the test of $\beta = 0$ and the interval.

```
## controls must be numeric columns, built in the packages chunk
ar <- AR_test(data = card, Y = "lwage", D = "education", Z = "near4",
              controls = c("age", "age2", "black", "smsa_yes", "south_yes"))
ar$Fstat      ## the AR test of beta = 0
#>           F      df1      df2      p
#>    4.138    1.000 3008.000    0.042
ar$ci         ## the 95% AR confidence set
#> [1] 0.0042 0.2561

## the usual 95% CI, estimate +/- 1.96 robust SE, for comparison
b_iv <- coef(m_iv)["education"]
se_iv <- sqrt(diag(vcovHC(m_iv, type = "HC1")))[["education"]]
round(c(estimate = unname(b_iv), lower = unname(b_iv - 1.96 * se_iv),
        upper = unname(b_iv + 1.96 * se_iv)), 4)
#> estimate      lower      upper
#>  0.0936   -0.0027   0.1899
```

How to read it:

- **The test of $\beta = 0$.** At $\beta_0 = 0$, $Y - \beta_0 D$ is just Y , so the AR test of “no effect” is the reduced-form test: $F = 4.14$ is essentially the squared robust t of the reduced form above (2.03^2). It needs no assumption about instrument strength.
- **The interval.** The 95% AR set is $[0.004, 0.256]$: the returns to a year of schooling, from 0.4% to 25.6%, that the data cannot reject at the 5% level. With a first-stage F of about 10, the instrument is weak by modern standards, so this is the interval to trust and report.
- **Compared with the usual 95% CI.** The usual interval, $[-0.003, 0.190]$, is symmetric around the estimate and just includes zero; at this F its 95% is not guaranteed. The AR set excludes zero and stretches further on the high side.

`ivDiag::ivDiag()` reports OLS, 2SLS, several first-stage F statistics, and the AR test in one call; it bootstraps by default, so it takes a minute.

6 Weak instruments, by simulation

The cleanest way to see the danger is to build a world where you know the answer. We use the lecture's design: $n = 500$, true effect 1, and strong endogeneity (the errors in D and Y correlate at 0.9), so OLS converges to 1.9.

```
set.seed(723)

one_draw <- function(pi, n = 500, beta = 1) {
  z <- rnorm(n); v <- rnorm(n)
  e <- 0.9 * v + sqrt(1 - 0.81) * rnorm(n)  ## error in Y, correlated with v
  d <- pi * z + v                          ## pi = instrument strength
  y <- beta * d + e                        ## TRUE effect is 1
  m <- ivreg(y ~ d | z)
  ## what the printed output shows: estimate, SE, first-stage F
  ## plus whether the AR test (at the truth) rejects the truth
  c(est = unname(coef(m)["d"]), se = sqrt(vcov(m)["d", "d"]),
    F   = unname(summary(lm(d ~ z))$fstatistic[1]),
    ar_rejects_truth = summary(lm(I(y - beta * d) ~ z))$coefficients["z", "t
    value"]^2 >
      qchisq(0.95, 1),
    ols = unname(coef(lm(y ~ d))["d"]))
}

strengths <- c(strong = 0.45, moderate = 0.11, weak = 0.06)
res <- lapply(strengths, function(p) as.data.frame(t(replicate(1000,
one_draw(p)))))

## the sampling distribution, summarized
sapply(res, function(r) c(
  median_F = median(r$F),
  median_IV = median(r$est),
  IQR_IV = IQR(r$est),
  median_SE = median(r$se),
  mean_OLS = mean(r$ols)
)) |> round(3)

#>           strong moderate  weak
#> median_F  101.521      6.127 1.689
#> median_IV   1.006      1.022 1.143
#> IQR_IV      0.133      0.530 0.970
#> median_SE   0.098      0.395 0.770
#> mean_OLS    1.748      1.889 1.897
```

```
## stack the three simulations into one data frame for ggplot
bind_rows(lapply(names(res), function(nm)
  tibble::tibble(est = res[[nm]]$est, strength = nm))) |>
filter(est > -3, est < 5) |>
ggplot(aes(est)) +
  geom_histogram(bins = 60, fill = "navy", alpha = .7) +
  geom_vline(xintercept = 1, linetype = 2) +
  geom_vline(xintercept = 1.9, linetype = 3) +
  facet_wrap(~ factor(strength, levels = c("strong", "moderate", "weak")),
    scales = "free_y") +
  labs(x = expression(hat(beta)[2*SLS]), y = "count") +
  theme_minimal(base_size = 9)
```

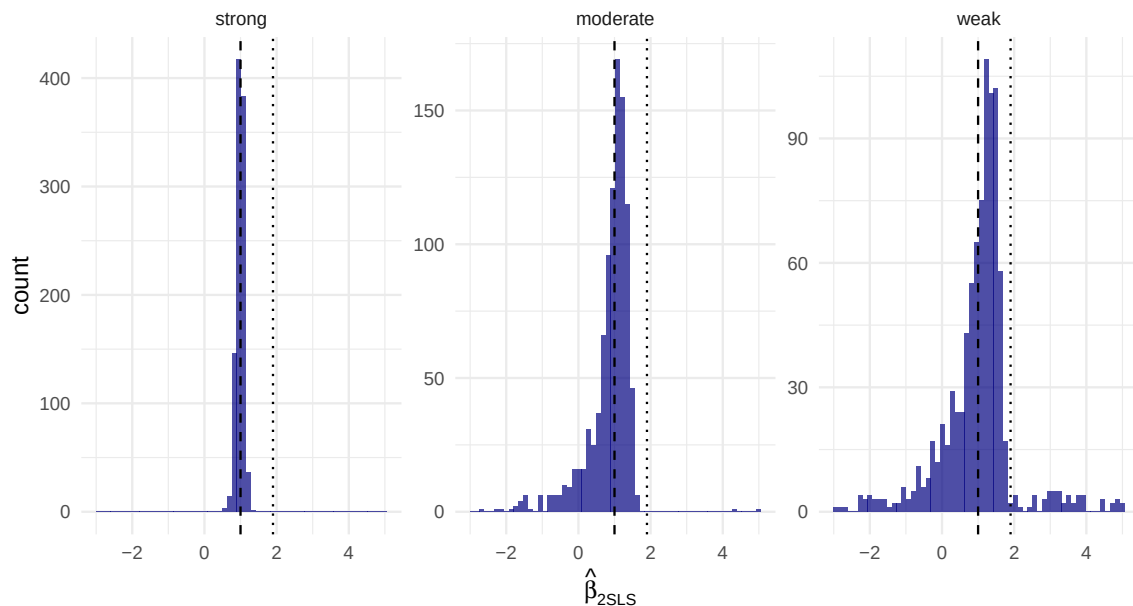


Figure 1: Sampling distribution of the 2SLS estimate at three instrument strengths, truncated to $[-3, 5]$ for display. The truth is 1 (dashed), the OLS limit 1.9 (dotted).

The strong instrument gives a tight, normal distribution centred on 1. As the instrument weakens, the distribution spreads out, grows heavy tails, and leans toward the OLS limit.

6.1 What the printed output does

A researcher never sees the histogram. They see one draw: an estimate, a standard error, and the interval $\hat{\beta} \pm 1.96 \widehat{SE}$. How often is that interval right?

```
read_out <- function(r) {
  covers <- abs(r$est - 1) <= 1.96 * r$se ## interval contains the truth
  signif <- abs(r$est / r$se) > 1.96      ## "significant" at 5%
```

```

c(CI_covers_truth      = mean(covers),
  share_significant    = mean(signif),
  CI_covers_when_signif = mean(covers[signif]),
  median_IV_when_signif = median(r$est[signif]),
  AR_covers_truth      = mean(!r$ar_rejects_truth))
}
sapply(res, read_out) |> round(2)
#>               strong moderate weak
#> CI_covers_truth      0.94      0.90 0.87
#> share_significant    1.00      0.60 0.42
#> CI_covers_when_signif 0.94      0.84 0.69
#> median_IV_when_signif 1.01      1.18 1.41
#> AR_covers_truth      0.94      0.95 0.95

```

Read the weak column:

- **Most weak draws look honest.** The typical draw has a large standard error, so its interval is wide and covers 1, and the estimate is not significant (58% of weak draws). That is why overall coverage of the usual CI is still 87%: a weak instrument usually prints a noisy, insignificant estimate.
- **The draws that look best are the worst.** Among the significant results, the ones a paper would report, the interval covers the truth only 69% of the time (84% at $F \approx 6$), and their median estimate is 1.41, not 1. The significant draws are the ones that drifted toward OLS, and the printed output cannot tell you which draw you have.
- **AR covers 95% in every column**, because it never divides by the first stage. This is why the diagnostics section tests with AR rather than reading the t -statistic.

7 Who are the compliers?

The LATE applies to people whose schooling was changed by college proximity. We cannot name them, but we can describe them.

```

## complier share = first stage on a binary treatment
card2 <- card |> mutate(some_college = as.integer(education >= 13))
fs_bin <- lm(some_college ~ near4 + age + I(age^2) + ethnicity + smsa + south,
             data = card2)
round(coef(summary(fs_bin))["near4", ], 4)
#>   Estimate Std. Error   t value   Pr(>|t|)
#>    0.0599    0.0205    2.9211    0.0035

```

About 6 percent of the sample are compliers on this definition — that is, roughly 6 percent attend some college *because* they grew up near one.

```

## Abadie's kappa weights let us describe complier characteristics;
## pz, the probability of growing up near a college given the controls,
## makes the weights valid when near4 is random only given the controls
pz <- fitted(glm(near4 ~ age + I(age^2) + ethnicity + smsa + south,
                 data = card2, family = binomial))
kappa <- 1 - card2$some_college * (1 - card2$near4) / (1 - pz) -
  (1 - card2$some_college) * card2$near4 / pz

## kappa-weighted mean of x = the compliers' mean of x
profile <- function(x) c(all = mean(x), compliers = sum(kappa * x) /
sum(kappa))
round(rbind(
  father_educ = profile(card2$feducation |>
tidyr::replace_na(mean(card2$feducation, na.rm = TRUE))),
  south = profile(card2$south_yes),
  smsa = profile(card2$smsa_yes),
  black = profile(card2$black)
), 3)
#>
#>           all compliers
#> father_educ 9.989      12.331
#> south       0.404      0.177
#> smsa        0.713      0.932
#> black       0.234      0.129

```

! Important

The complier profile is the interpretive heart of an IV paper. If the compliers are disproportionately from low-education families, then the LATE is a return to college **for people at the margin of attending** — which is often the policy-relevant group, but is emphatically not the ATE.

7.1 The same profile with ivdesc

`ivdesc()` (Marbach and Hangartner 2020) computes the mean of a covariate for compliers (`co`), never-takers (`nt`), and always-takers (`at`), with standard errors, from a binary treatment and a binary instrument.

```

## X = covariate to profile, D = binary treatment, Z = binary instrument
with(card2, ivdesc(X = black, D = some_college, Z = near4, boot = FALSE))
#>
#>
#> |group |      mu| mu_se|      pi| pi_se|
#> |:-----|-----:|-----:|-----:|-----:|

```

```
#> |sample | 0.2336| 0.0077| 1.0000| 0.0000|
#> |co      | 0.2102| 0.0649| 0.1219| 0.0194|
#> |nt      | 0.3002| 0.0150| 0.4559| 0.0110|
#> |at      | 0.1683| 0.0186| 0.4222| 0.0160|
#>
#> Balance test: H0: E[X|Z=0]=E[X|Z=1]
#> Pr(|T| > |t|) = 0.0000698
```

`ivdesc()` assumes the instrument is randomly assigned **without** conditioning on covariates. Its balance test at the bottom asks whether the covariate differs by Z , and here it rejects: Black respondents are less likely to have grown up near a college. So this table is the profile for a design in which `near4` was a lottery, and the covariate-adjusted κ above is the right one for Card's design. The same goes for its complier share, 12 percent here against 6 percent with the controls. Use `ivdesc()` when Z is randomized (a lottery, an experiment's encouragement), and the hand-built κ when it is random only given controls.

8 Exercises

1. **Reduced form first.** Suppose a referee reports only the 2SLS estimate and its t -statistic. Compute the reduced form and first stage and write two sentences a referee could use to evaluate whether the design is credible.
2. **A placebo.** Growing up near a college should not affect variables determined before college attendance. Regress `feducation`, `meducation`, and `library14` on `near4` with the same controls. Do any of them move? What would it mean for exclusion if they did?
3. **Anderson–Rubin by hand.** Write a function that, for a grid of β_0 values, regresses $Y - \beta_0 D$ on the instrument and controls and records the robust p -value on the instrument. Plot it, read off the 95% AR confidence set, and check it against `ar$ci`.
4. **Bias amplification with weak instruments.** In `one_draw()`, add a small direct effect of Z on Y (violating exclusion by, say, 0.05). Show that the resulting bias in 2SLS grows as the instrument weakens, and compare it with the OLS bias.

9 Session information

```
## R version and the packages this lab loaded
c(R = R.version.string, supply(sessionInfo())$otherPkgs, \(p)
as.character(p$Version)))
#>                                     R                               ivdesc
#> "R version 4.5.2 (2025-10-31)"      "1.1.2"
#>                                ivDiag                             lmtest
#>                                "1.0.6"                             "0.9-40"
```



```
#>          zoo          sandwich
#>      "1.8-15"      "3.1-1"
#>      fixest      ivreg
#>      "0.14.2"      "0.6-8"
#>      ggplot2      dplyr
#>      "4.0.2"      "1.2.0"
```